# TDAE: Tensored Deep Autoencoder for Classification of Hyperspectral Images

Changda Xing, Meiling Wang, Xuesong Wang, *Member, IEEE*, and Yuhu Cheng, *Member, IEEE*

***Abstract*— Deep learning has achieved outstanding success in the hyperspectral image (HSI) classification task. Almost all the current deep learning methods are used to conduct classification predictions by leveraging the output features from the deepest layer, which generally ignore the attention to multilayer outputs, so that the capability of hierarchical representation is limited. To remedy such deficiency, in this article, we propose to build a novel deep network form, called tensored deep autoencoder network (TDAE), for HSI classification. For this method, the tensor decomposition constraint item is built and introduced into a deep autoencoders network with a fully connection layer. It not only achieves the integration of multilayer output features but also captures the structure information among outputs. By such way, the network's ability for hierarchical representation is significantly enhanced. Furthermore, to solve such built model, we further design an alternating update optimization scheme and obtain the desired feature forms. The features are further input into the fully connection layer to generate the label of the given HSI. Extensive experiments have been conducted to validate that the proposed TDAE method achieves more competitive performance compared with several state-of-the-art approaches.**

***Index Terms*— Deep learning, hierarchical representation, hyperspectral image (HSI) classification, multilayer outputs, tensored deep autoencoder (TDAE).**

## I. Introduction

HYPERSPECTRAL image (HSI) is a type of spectral imaging data that boast an extremely high spectral resolution [1]. These data are often collected using hyperspectral sensors located on various platforms (such as satellites and unmanned aerial vehicles) over a target area across numerous, continuous, and finely divided spectral bands, spanning the ultraviolet, visible, near-infrared, and mid-infrared regions of the electromagnetic spectrum [2], [3], [4]. With its specific characteristic of "integrating spectrum and image," HSI plays a significant role in various fields. Among HIS-related research problems, HSI classification is an important and widely research direction, which aims to categorize each pixel in the image into a specific category [5], [6].

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For HSI classification, the most direct and simple way is to treat hyperspectral pixels as discriminative features and use a classifier such as support vector machine (SVM) to yield labels [7]. However, this way has no ability to achieve the competitive robustness and stability, due to the utilization of a large amount of raw and redundant information. Representation learning (RL) [8], [9], [10] is an available manner to solve the above challenge. As a kind of powerful machine learning tools, RL has shone brightly in HSI classification, whose fundamental is to extract more compact and discriminative feature representations from raw hyperspectral data. These features can more effectively capture the spectral and spatial information of objects, thereby enhancing the accuracy and efficiency of classification. Sparse representation and low-rank representation are two types of common RL forms for solving the problem of HSI classification. For sparse representation, it is assumed that an unknown sample is approximately represented by the combination of a few training samples from the entire dictionary, whereas the corresponding sparse vector encodes the class information implicitly [11]. Scholars have usually introduced L1-norm-regularization [12], Laplacian regularized Lasso [13], group sparse coding [14], and joint sparsity [15] to construct the frameworks of HSI classification. For low-rank representation, a particular form under the lowest rank constraint is built to reduce redundancy and noise in HSI data by discovering global structure of data [16], which refers to the underlying patterns that can be represented with fewer information or features. For instance, the low-rank discriminative least-squares regression model (LRDLSR) [17] and the diagonalized low-rank learning (DLRL) model [18] classwisely impose the low-rank constraint to mine structural characteristics for improving the classification performance.

As a kind of powerful learning manners, deep learning [19], [20], [21], [22], [23], [24], [25], [26], [27], [28], [29] is able to automatically extract multilevel, multiscale, and nonlinear features from HSI data without the need for manually designing complex feature extraction schemes, which can enable efficient classification performance. To the best of our knowledge, the pioneering introduction of deep learning for HSI classification was documented in [19], where stacked autoencoders (SAEs) were leveraged to extract discriminative deep-level features and a logistic regression classifier was used to classify these deep features and determine the label map of the given HSI. Zhou et al. [21] also proposed a representative framework named as compact and discriminative SAE (CDSAE) was

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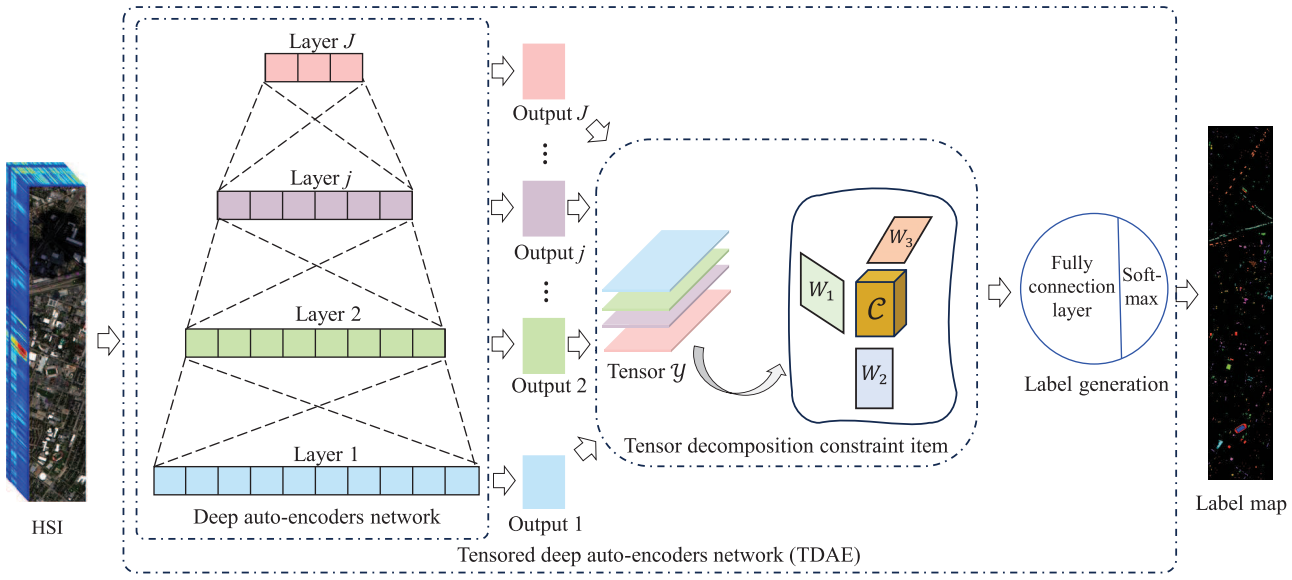



Fig. 1. Overview of the proposed TDAE method for HSI classification.

designed, which can learn discriminative low dimensional feature mappings and train an effective classifier progressively. In [22], a novel deep ring-blockwise autoencoder network (DRN) was proposed, which achieves good separability and discrimination of features by constructing a ring-block perception layer.

More popularly, the convolutional neural network (CNN) [30], [31], [32], [33], [34] has been widely introduced to deal with the problem of HSI classification. Li et al. [31] proposed to use a deep CNN to learn pixel-pair features to achieve more discriminative power for the great HSI classification results. The squeeze multibias network (SMBN) was proposed [32], which introduces an efficient squeeze convolution module to efficiently improve the running speed of the network by reducing the number of parameters, while the high classification accuracy is still maintained. More advancedly, Shen et al. [33] developed an efficient nonlocal module based-deep fully convolutional network (ENL-FCN) to incorporate the long-range contextual information for HSI classification. Xu et al. [34] proposed a robust self-ensembling network (RSEN) to address the classification task.

In recent several years, the transformer network (TN) [35], [36], [37], [38] has also been widely used for classification. For example, Graham et al. [36] designed a named LeViT method, which introduces the attention bias to integrate positional information in vision transformers for the purpose of classification. Yang et al. [37] proposed a hyperspectral image transformer (HiT) classification network by embedding convolution operations into the transformer structure to capture the subtle spectral discrepancies and convey the local spatial context information. In [38], the self-supervised masked image reconstruction was introduced to advance transformer models for the classification of hyperspectral remote sensing imagery.

We should note that to the best of our knowledge, almost all the above deep learning methods are used to conduct classification predictions by leveraging the output features from the deepest layer. However, it is necessary to emphasize here that although the deepest layer yields more valuable and robust feature information, the attention to multilayer outputs is generally neglected, which decreases the capability of neural networks for hierarchical RL. In other words, current structures of neural networks have no ability to capture rich feature information from different perception levels. They often overly focus on the output of the deepest layer during their design and implementation, lacking comprehensive attention and effective utilization of multilayer outputs.

To remedy the above-discussed deficiency, we propose to build a novel deep network framework, called tensored deep autoencoder (TDAE) network, for the classification of HSIs. The overview of the proposed TDAE method is shown in Fig. 1. More concretely, we first introduce a deep autoencoders' network with a fully connection layer as the basic framework. Second, the tensor decomposition constraint item is built and introduced into the basic framework, which not only achieves the integration of multilayer output features but also captures the structure information among outputs. By such way, the network's ability for hierarchical representation is significantly enhanced. Third, to solve the constructed model, we theoretically explore the relationship between the tensor decomposition and matrix form, and based on this, we further design an alternating update optimization scheme. Finally, once the optimal solutions are obtained, the desired feature forms will be extracted and put into the fully connection layer to generate the label of the given HSI. The design details of this TDAE method are shown in Section II. Extensive experimental results have been given to validate the effectiveness and superiority of the proposed TDAE method.

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The main contributions of this article are summarized into the following four points. *First*, we construct a novel deep network framework, i.e., TDAE, which can achieve the great capability of hierarchical RL to improve the performance of HSI classification. *Second*, we for the first time use the tensor idea to rebuild the deep autoencoders' network form, which not only achieves the integration of multilayer output features but also captures the structure information among outputs. *Third*, we design an alternating update optimization scheme to solve the built TDAE model. *Fourth*, we experimentally explore the performance of the TDAE method, with the results demonstrating its effectiveness and superiority.

The rest of this article is organized below. We introduce the design details of the proposed TDAE method in Section II. The experimental exploration is conducted in Section III. The conclusion is finally presented in Section IV.

## II. Methodology

In this section, we will present the design details of the proposed TDAE method for HSI classification. Here, we denote the input HSI by $\boldsymbol{X} \in \mathbb{R}^{S\times P\times L}$ and the corresponding 2-D matrix form by $X \in \mathbb{R}^{Q\times L}$, where $Q = S \times P$ is the spatial resolution and $L$ is the spectral resolution.

### A. Motivation Exploration

Lots of works [19], [20], [21], [22], [23], [24], [25], [26], [27], [28], [29] have already clearly clarified that deep learning can automatically extract high-level abstract features from the original HSI data and realize the high-performance of HSI classification. Most of them are intricately structured with neural networks, such as CNNs and DAEs, which are composed of several hidden layers (often more than two layers) to seek the specific functions reflecting the mapping from the input (i.e., HSI data) to the output (i.e., discriminative features or categorical labels). In general, most current relevant methods use the deepest layer features to predict classification results, since more complex and advanced semantic information as well as rich expressive capabilities are included in such layer. It should be noted that although the deepest layer can yield more valuable and robust feature information, the attention to multilayer outputs is generally neglected, which decreases the capability of neural networks for hierarchical RL. *That is*, current structures of neural networks have no ability to capture rich feature information from different perception levels. They often overly focus on the output of the deepest layer during their design and implementation, lacking comprehensive attention and effective utilization of multilayer outputs [23], [39]. Consequently, current neural-network-based methods for HSI classification tend to neglect features (such as shallow features) from other levels beyond the deepest layer. As can be known, shallow features are yielded from the first few layers of networks, which has the competitive advantage over deep features in representation of detail information from the input data [40]. Many literature such as [41] imply that these shallow features are also less sensitive to small changes in input data, and therefore have a certain degree of stability. Though shallow features are not as abstract and global as deep features, they provide necessary supplement for deep features. In other words, the neglect of shallow features seems to be a kind of information loss, which limits the performance on HSI classification. It is very possible to be highly valuable and meaningful for HSI classification to integrate both deep and shallow features together. Hence, the purpose of our work is to seek a way to concurrently use both deep and shallow features for classifying HSIs, which is the *motivation* of this work.

### B. Theoretical Construction

Among numerous available deep network models, the DAE form is one of the earliest and very influential models used in the HSI classification task. This form is a network model that learns effective representations of data through unsupervised learning. The core idea is that through a multilayer structure of encoders and decoders, the DAEs can learn a compact representation of the input HSI data and reconstruct the original HSI data from this compact representation through the decoding operation. Given the input HSI data $X$, the DAE form can be represented as follows:

$$\begin{aligned} &\underset{V_j,U_j,Y_j}{\arg\min} \sum_{j=1}^{J}\left\|X_j - U_jY_j\right\|_F^2 + \sum_{j=1}^{J}\lambda\left\|Y_j\right\|_1 \\ &\text{s.t., } Y_j = V_jX_j, \quad j = 1, 2, \ldots, J \end{aligned} \tag{1}$$

where $\lambda$ denotes the regularization parameter; $V_j$, $U_j$, $X_j$, and $Y_j$ are the encoding matrix, the decoding matrix, the input, and the output at the $j$th layer of the network, where $X_j = Y_{j-1}$ $(j \geq 2)$ and $X_1 = X$. Almost all the existing methods based on (1) or its extended forms take the output of the deepest layer as features and then yield the classification results by a prediction layer such as soft-max. It is noted that like the descriptions on the motivation exploration above, these methods lack the comprehensive attention and effective utilization of multilevel features (i.e., multilayer outputs), so that they tend to ignore the features from other levels beyond the deepest layer.

From the perspective of algorithmic implementation, the most direct way to simultaneously use multilevel features is to concatenate such multilevel features in sequence and then input the concatenated features into the prediction layer. However, this simple way not only suffers from a substantial amount of redundancy in the concatenated features but also fails to capture the structural information across hierarchical representations. For the sake of comprehensively using multilevel features for HSI classification, we consider to stack these multilevel features into a tensor $\mathcal{Y} \in \mathbb{R}^{Q\times L'\times J}$ in this work and thus perform tensorization on the above formula (1). Here, due to numerous redundancy and structural information included in $\mathcal{Y}$, we need to introduce the concept of tensor decomposition to reduce redundancy while preserving the structural information of hierarchical representations. Based on these discussions, the

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TDAE model for classification of HSI can be written into

$$\begin{aligned}\underset{V_j,U_j,Y_j,\mathcal{Y},\mathcal{C},W_1,W_2,W_3}{\arg\min}\ & \sum_{j=1}^{J}\|X_j-U_jY_j\|_F^2+\sum_{j=1}^{J}\lambda\|Y_j\|_1\\ &+\alpha\|\mathcal{Y}-\mathcal{C}\times W_1\times W_2\times W_3\|\\ \text{s.t., } & Y_j=V_jX_j,\quad j=1,2,\ldots,J\\ & \mathcal{Y}=\text{Cat}(Y_1,Y_2,\ldots,Y_J)\end{aligned}\tag{2}$$

where $\alpha$ is a coupling parameter used to control the significance of the tensor decomposition item; $\text{Cat}(\cdot)$ is an operator that catenates all $Y_j(j=1,\ldots,J)$ along layers as the tensor $\mathcal{Y}$. Note that since the dimensions of different $Y_j$ may be various, it is necessary to use the maximum dimension as the benchmark, and the other smaller dimensions should be padded with zeros to achieve consistent dimensions. $\mathcal{C}\in\mathbb{R}^{Q''\times L''\times J''}$ denotes the core tensor of $\mathcal{Y}$, where $Q\gg Q''$, $L'\gg L''$, and $J\gg J''$; $W_1\in\mathbb{R}^{Q\times Q''}$; $W_2\in\mathbb{R}^{L'\times L''}$; and $W_3\in\mathbb{R}^{J\times J''}$. As can be directly found from (2), the outputs of all the layers are constructed as a tensor; meanwhile, the tensor constraint also generates a backpropagation effect on the encoding network. This can achieve the optimal hierarchical representations once (2) is optimally solved.

### C. Applicable Optimization

There exist both matrix operations and tensor operations simultaneously in (2), and therefore, it is extremely challenging to directly solve this formula. It is best to unify the form of operations.

*Theorem 1:* Given a tensor $\mathcal{Y}\in\mathbb{R}^{Q\times L'\times J}$, the corresponding Tucker-2-based tensor decomposition is equal to $J$ joint matrix decompositions.

*Proof:* For the given tensor $\mathcal{Y}\in\mathbb{R}^{Q\times L'\times J}$, the corresponding Tucker decomposition can be expressed as

$$\mathcal{Y}\approx\mathcal{C}\times W_1\times W_2\times W_3\tag{3}$$

where core tensor $\mathcal{C}$ has the size of $Q''\times L''\times J''$ ($Q\gg Q''$, $L'\gg L''$, and $J\gg J''$). The factor matrices $W_1\in\mathbb{R}^{Q\times Q''}$; $W_2\in\mathbb{R}^{L'\times L''}$; $W_3\in\mathbb{R}^{J\times J''}$ are the potential principal components of the expanded matrix under different modes. Here, the factor matrix corresponding to the third mode is constrained to be an identity matrix, and thus the optimization problem of Tucker-2-based tensor decomposition is here given below

$$\underset{\mathcal{S},W_1',W_2'}{\arg\min}\ \|\mathcal{Y}-\mathcal{S}\times W_1'\times W_2'\|_F^2\tag{4}$$

where $\mathcal{S}\in\mathbb{R}^{Q''\times L''\times J''}$ denotes the core tensor under the factor matrices $W_1'\in\mathbb{R}^{Q\times Q''}$ and $W_2'\in\mathbb{R}^{L'\times L''}$. Since $\mathcal{Y}$ is the catenation of all $Y_j(j=1,\ldots,J)$ along the layer direction, the Tucker decomposition in (4) is equal to the following matrix decomposition:

$$\underset{W_1',F_j,W_2'}{\arg\min}\ \sum_{j=1}^{J}\|Y_j-W_1'\times F_j\times W_2'^T\|_F^2\tag{5}$$

where $F_j$ is the intrinsic matrix of $Y_j$. $W_1'$ and $W_2'$ are the corresponding factor matrix. Formula (5) aims to obtain $J$ joint matrix decompositions. The visual presentation is seen in Fig. 2.

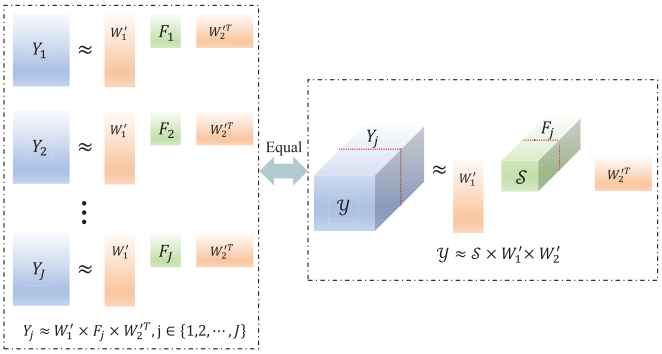



Fig. 2. Visual presentation of the Tucker-2 tensor decomposition is equal to $J$ joint matrix decompositions.

According to Theorem 1, the TDAE model in (2) can be written as the following form without tensor factors:

$$\begin{aligned}\underset{V_j,U_j,Y_j,F_j,W_1',W_2'}{\arg\min}\ & \sum_{j=1}^{J}\|X_j-U_jY_j\|_F^2+\sum_{j=1}^{J}\lambda\|Y_j\|_1\\ &+\alpha\sum_{j=1}^{J}\|Y_j-W_1'F_jW_2'^T\|_F^2\\ \text{s.t., } & Y_j=V_jX_j,\quad j=1,2,\ldots,J.\end{aligned}\tag{6}$$

This optimization problem is further simplified into the unstrained form below

$$\begin{aligned}\underset{V_j,U_j,Y_j,F_j,W_1',W_2'}{\arg\min}\ & \sum_{j=1}^{J}\Big(\|X_j-U_jY_j\|_F^2+\lambda\|Y_j\|_1\\ &+\alpha\|Y_j-W_1'F_jW_2'^T\|_F^2\\ &+\gamma\|Y_j-V_jX_j\|_F^2\Big)\end{aligned}\tag{7}$$

where $\gamma$ is the coupling parameter. As can be seen, the objective function in (7) is a multivariable nonconvex optimization problem, which is difficult to be directly solved. However, this objective function is convex for single variable when other variables are fixed. Thus, we adopt the alternating update optimization scheme to solve the problem. An auxiliary variable $A_j=Y_j$ is here introduced, and (7) can be expressed as

$$\begin{aligned}\underset{V_j,U_j,Y_j,F_j,W_1',W_2',A_j}{\arg\min}\ & \sum_{j=1}^{J}\Big(\|X_j-U_jY_j\|_F^2+\lambda\|A_j\|_1\\ &+\alpha\|Y_j-W_1'F_jW_2'^T\|_F^2\\ &+\gamma\|Y_j-V_jX_j\|_F^2\\ &+\beta\|A_j-Y_j\|_F^2\Big)\end{aligned}\tag{8}$$

where $\beta$ denotes the coupling parameter. Thus, the procedure of the alternating update optimization scheme is shown below.

*1) Optimization of $V_j$:* We keep the variables $U_j$, $Y_j$, $F_j$, $W_1'$, $W_2'$, and $A_j$ fixed, and $V_j$ can be obtained by optimizing

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the following subproblem:

$$\arg\min_{V_j} \sum_{j=1}^{J} \left\| Y_j - V_j X_j \right\|_F^2. \tag{9}$$

This formula has the closed-form solution below

$$V_j = \left(Y_j X_j^T\right)\left(X_j X_j^T\right)^{-1}. \tag{10}$$

*2) Optimization of $U_j$:* Fixing $V_j$, $Y_j$, $F_j$, $W_1'$, $W_2'$, and $A_j$, $U_j$ is acquired by

$$\arg\min_{U_j} \sum_{j=1}^{J} \left\| X_j - U_j Y_j \right\|_F^2. \tag{11}$$

Like (9), (11) also has the following solution:

$$U_j = \left(X_j Y_j^T\right)\left(Y_j Y_j^T\right)^{-1}. \tag{12}$$

*3) Optimization of $Y_j$:* Keeping $V_j$, $U_j$, $F_j$, $W_1'$, $W_2'$, and $A_j$ fixed, $Y_j$ can be updated via

$$\arg\min_{Y_j} \sum_{j=1}^{J} \Big( \left\| X_j - U_j Y_j \right\|_F^2 + \alpha \left\| Y_j - W_1' F_j W_2'^T \right\|_F^2 + \gamma \left\| Y_j - V_j X_j \right\|_F^2 + \beta \left\| A_j - Y_j \right\|_F^2 \Big). \tag{13}$$

The solution of (13) is

$$Y_j = \left(U_j^T U_j + I\right)^{-1} \times \left(U_j^T X_j + \alpha W_1' F_j W_2'^T + \gamma V_j X_j + \beta A_j\right). \tag{14}$$

*4) Optimization of $F_j$:* Fixing $V_j$, $U_j$, $Y_j$, $W_1'$, $W_2'$, and $A_j$, the variable $F_j$ is optimally updated through

$$\arg\min_{F_j} \sum_{j=1}^{J} \left\| Y_j - W_1' F_j W_2'^T \right\|_F^2. \tag{15}$$

As can be seen, directly solving the problem will include two inverse operations, i.e., $(W_1'^T W_1')^{-1}$ and $(W_2'^T W_2')^{-1}$, which extremely limit the computational efficiency. However, problem (15) is quadratic and its unique solution is equivalent to compute the general Sylvester matrix equation

$$W_1'^T W_1' F_j W_2'^T W_2' = W_1'^T Y_j W_2'. \tag{16}$$

The conjugate gradient (CG) [42] is used to solve (16). For the CG scheme, we should know: *first*, the system matrix associated with (16) is symmetric and positive definite, which is a necessary and sufficient condition to directly apply CG. *Second*, the step of the heaviest computation in applying CG is the multiplication of the system matrix times a vector, whose computation consumption is much less than that of inverse operations. *Third*, CG converges in only a few iterations. The convergence of CG is linear and depends on the ratio between the largest and the smallest eigenvalues of the system matrix associated with (16) [43].

*5) Optimization of $W_1'$:* We set $V_j$, $U_j$, $Y_j$, $F_j$, $W_2'$, and $A_j$ to be fixed, and the optimal $W_1'$ can be obtained by

$$\arg\min_{W_1'} \sum_{j=1}^{J} \left\| Y_j - W_1' F_j W_2'^T \right\|_F^2 \tag{17}$$

whose solution is described as

$$W_1' = \left( \sum_{j=1}^{J} Y_j \left(F_j W_2'^T\right)^T \right) \left( \sum_{j=1}^{J} \left(F_j W_2'^T\right)\left(F_j W_2'^T\right)^T \right)^{-1}. \tag{18}$$

*6) Optimization of $W_2'$:* Similar to the optimization $W_1'$, $W_2'$ can be optimally described as

$$W_2'^T = \left( \sum_{j=1}^{J} \left(W_1' F_j\right)^T \left(W_1' F_j\right) \right)^{-1} \left( \sum_{j=1}^{J} \left(W_1' F_j\right)^T Y_j \right). \tag{19}$$

*7) Optimization of $A_j$:* $V_j$, $U_j$, $Y_j$, $F_j$, $W_1'$, and $W_2'$ are fixed, and the optimization of $A_j$ is achieved via

$$\arg\min_{A_j} \sum_{j=1}^{J} \left( \lambda \left\| A_j \right\|_1 + \beta \left\| A_j - Y_j \right\|_F^2 \right) \tag{20}$$

which yields the closed-form solution below

$$A_j = \text{soft}\left(Y_j, \frac{\lambda}{2\beta}\right) \tag{21}$$

where $\text{soft}(x, y) = \text{sign}(x) \times \max(|x| - y, 0)$.

The optimal variables will be obtained until convergence. The convergent condition lies in the following forms:

$$\begin{aligned} \left\| X_j - U_j Y_j \right\|_F^2 &< \epsilon_1 \\ \left\| Y_j - W_1' F_j W_2'^T \right\|_F^2 &< \epsilon_2 \\ \left\| Y_j - V_j X_j \right\|_F^2 &< \epsilon_3 \end{aligned} \tag{22}$$

where $\epsilon_1$, $\epsilon_2$, and $\epsilon_3$ are three small constants. Here, a plot is provided to visually show the evolution of the cost function of our TDAE in (8) to clearly observe the optimization process in Fig. 3. From this figure, it can be easily seen that overall, the value of the cost function decreases to a steady state (i.e., convergence point) with the increase in iteration times, which only spends about 25 iterations.

Algorithm 1 also summarizes the optimal estimation of $V_j$, $U_j$, $Y_j$, $F_j$, $W_1'$, and $W_2'$.

### D. Feature Representation and Label Generation

When the optimization procedure is completed, the solutions $F_j$ and $W_1'$ are obtained. As can be known from the previous descriptions, $F_j$ is the intrinsic component of the output $Y_j$ of each layer, and $W_1'$ denotes the transformation projection of $Y_j$ along the spectrum or feature direction. Considering the hierarchical representation, the final feature form $F$ can be written as

$$F = \sum_{j=1}^{J} W_1' F_j. \tag{23}$$

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**Algorithm 1** Optimization Procedure for Problem (8)

**input** : HSI data $X$; Parameters $\alpha$, $\lambda$, $\gamma$, and $\beta$
**output**: $V_j$, $U_j$, $Y_j$, $F_j$, $W_1'$, $W_2'$

**while** *not converged* **do**
  **Step 1** Update $V_j$ via (10);
  **Step 2** Update $U_j$ via (12);
  **Step 3** Update $Y_j$ via (14);
  **Step 4** Update $F_j$ via (16);
  **Step 5** Update $W_1'$ via (18);
  **Step 6** Update $W_2'$ via (19);
  **Step 7** Update $A_j$ via (21);
**end while**

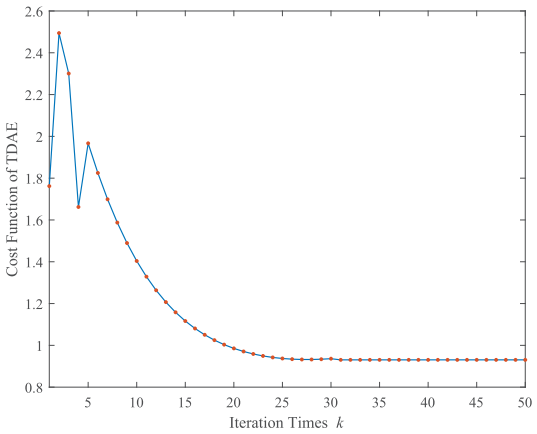



Fig. 3. Evolutionary curve of TDAE in (8) with iteration times $k$.

As shown in (23), the final feature representation $F$ is equipped with the multilayer outputs, which combines the properties of both shallow features and deep features.

To yield the label map of the input HSI data, the feature output channel of the network is linked to a fully connected layer based on soft-max. That is, the final features are input into a fully connected layer to further generate the label of HSI.

## III. Experiment

### A. Experimental Preparation

*1) Datasets:* We used the Salinas, Trento, and Houston datasets to assess the performance of the proposed TDAE method on the task of HSI classification. The Salinas data[1] were collected using the 224-band airborne/visible infrared imaging spectrometer (AVIRIS) sensor over Salinas Valley at California, which has 16 classes. To achieve the convenient analysis, we removed 20 water absorption bands and down-sampled spatial resolution to obtain the preprocessed data with 512 × 217 × 204. The Trento dataset[2] was captured over a rural region in the south of Trento, Italy. These HSI data were acquired through the AISA Eagle system for hyperspectral imaging Eagle sensor, which is equipped with 63 spectral bands covering the range from 0.42 $\mu$m to 0.99 $\mu$m and 600 × 600 spatial resolution at the 1-m interval, where six classes are included. The Houston dataset[3] was obtained the area of the campus of the University of Houston and the neighboring urban, which contains 15 labeled classes and 349 × 1905 × 144 size. It should be noted that for convenient observation of experimental results, in the following experiments, we, respectively, select 5%, 1%, and 1% ratio from the Salinas, Trento, and Houston datasets as training samples, and the rests are testing samples. Selecting these portion values is that the comparison results of different methods are easily and obviously observed under the 5%, 1%, and 1% portions for Salinas, Trento, and Houston.

*2) Quantitative Metrics:* To quantitatively evaluate the experimental results, we introduced three typical evaluation metrics, namely, overall accuracy (OA), average accuracy (AA), and kappa coefficient (Kappa). OA is the percentage of pixels properly categorized. AA is the average percentage of correctly classified pixels for each class. Kappa corrects the correct percentage of classified pixels.

### B. Parameter Analysis

In this section, we will conduct experiments to analyze the impact of parameters (including $\alpha$, $\lambda$, $\gamma$, and $\beta$) on the performance of the proposed TDAE method. It is known that $\alpha$ is a coupling parameter used to control the importance of tensor decomposition item. The $\lambda$ denotes the regularization parameter, which controls the sparsity of the output of each layer. The $\gamma$ is used to control the weight of the constraint $Y_j = V_j X_j$, which reflects the property of the decoding of the TDAE method. The $\beta$ is introduced during the optimization stage, which tunes the significance of the usage of auxiliary $A_j = Y_j$. Fig. 4 is adopted to present the classification performance of the proposed TDAE method influenced by different values of $\alpha$, $\lambda$, $\gamma$, and $\beta$ on the Salinas, Trento, and Houston datasets, where the OA metric is here used to evaluate the results. We use the grid search strategy to iterate over the parameter-value range of $[10^{-3}, 10^{-2}, 10^{-1}, 10^{0}, 10^{1}, 10^{2}]$.

*1) Analysis of Parameter $\alpha$:* Fig. 4(a) shows the classification performance (in OA) of the TDAE method affected by $\alpha$. As can be found from this figure, the classification performance achieves the best value at $\alpha = 10^0$ or $10^1$ on the Salinas, Trento, and Houston datasets. When the value of parameter $\alpha$ is relatively small (i.e., $\alpha \leq 10^0$), the significance of tensor decomposition item in (2) is extremely decreasing down, so that the ability of hierarchical presentation is weakened. When the value of $\alpha$ is too large (i.e., $\alpha > 10^1$), the perception capability of deep networks will be weakened, which leads to a decrease in the ability of feature representation.

*2) Analysis of Parameter $\lambda$:* As shown in Fig. 4(b), the proposed MTFR method obtains the largest OA values at $\lambda = 10^{-1}$ on the three datasets. When $\lambda < 10^{-1}$, the performance of the MTFR method will deteriorate, since the

[1]https://www.ehu.eus/ccwintco/index.php?title = Hyperspectral_Remote_Sensing_Scenes

[2]https://drive.google.com/drive/folders/1HK3eL3loI4Wd-RFr1psLLmVLTVDLctGd

[3]http://www.grss-ieee.org/community/technical-committees/datafusion/2013-ieee-grss-data-fusion-contest

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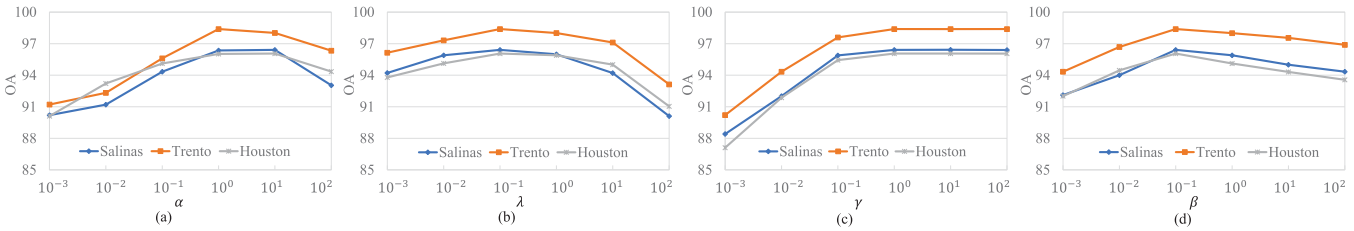



Fig. 4. Classification performance (in OA) of our proposed TDAE method affected by four parameters (i.e., $\alpha$, $\lambda$, $\gamma$, and $\beta$). (a) For $\alpha$. (b) For $\lambda$. (c) For $\gamma$. (d) For $\beta$.

TABLE I
METRIC RESULTS OF DIFFERENT METHODS ON THE SALINAS DATASET

| Categories | Classification Methods | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | RAW-SVM | LRDLSR | DRDL | CNN-PPN | CDSAE | LeViT | HiT | DRN | TDAE |
| Broccoli green weeds_1 | 100 | 91.04 | 83.80 | 99.20 | 99.80 | 97.40 | 99.81 | 99.80 | **100** |
| Broccoli green weeds_2 | 96.12 | 99.14 | 99.24 | 94.15 | **100** | 91.03 | 99.35 | 99.24 | 99.55 |
| Fallow | 58.45 | 90.02 | 99.80 | 99.80 | 99.18 | 98.37 | 94.89 | 99.80 | **99.82** |
| Fallow rough plow | 97.65 | 98.82 | 92.63 | 98.82 | **100** | **100** | 97.63 | 98.82 | 99.70 |
| Fallow smooth | 97.72 | 84.24 | 86.15 | 99.09 | 93.61 | 98.93 | 99.54 | 94.05 | **99.69** |
| Stubble | 98.69 | 94.65 | 99.59 | **100** | 96.96 | 98.58 | 92.28 | 98.48 | 99.69 |
| Celery | 78.16 | 98.77 | 99.89 | 99.78 | **99.89** | 94.49 | 99.66 | 99.66 | 99.65 |
| Grapes untrained | 57.46 | 63.89 | 76.07 | 73.18 | 87.73 | 78.15 | 86.62 | 84.00 | **95.25** |
| Soil vineyard develop | **99.87** | 95.09 | 97.80 | 97.73 | 99.16 | 98.12 | 98.77 | 98.57 | **99.87** |
| Corn senesced green weeds | 93.83 | 95.81 | 85.91 | 98.64 | 71.82 | 76.89 | 98.02 | 96.28 | **99.74** |
| Lettuce romaine 4 week | 95.86 | 94.74 | 91.32 | 93.96 | 93.58 | 84.53 | 95.85 | 76.98 | **99.61** |
| Lettuce romaine 5 week | 80.54 | 98.74 | 98.95 | **100** | 99.18 | 99.79 | **100** | **100** | **100** |
| Lettuce romaine 6 week | 98.68 | 98.68 | 94.27 | 98.23 | 96.48 | 51.98 | 98.23 | 98.67 | **99.54** |
| Lettuce romaine 7 week | 72.66 | 88.39 | 94.74 | 91.35 | 96.24 | **98.12** | 88.72 | 90.60 | 92.64 |
| Vineyard untrained | 75.99 | 80.40 | 78.88 | 75.03 | 79.22 | 78.83 | 73.74 | **84.94** | 84.87 |
| Vineyard vertical trellis | 68.34 | 48.64 | 88.36 | 87.19 | 96.58 | 96.80 | **98.86** | 98.40 | 91.51 |
| OA(%) | 81.67 | 84.92 | 88.73 | 89.46 | 91.88 | 88.44 | 92.16 | 93.03 | **96.41** |
| AA(%) | 85.63 | 88.82 | 91.71 | 94.05 | 94.34 | 90.13 | 95.12 | 94.89 | **97.57** |
| Kappa(%) | 79.46 | 83.16 | 87.44 | 87.49 | 90.93 | 87.11 | 91.25 | 92.24 | **96.00** |

sparsity becomes weaker at the excessively small $\lambda$ value level. On the contrary, when $\lambda > 10^{-1}$, the classification performance tends to decline due to the weakening of the network's perception ability caused by overemphasizing sparsity. Therefore, $\lambda = 10^{-1}$ is considered as the optimal selection of the parameter value.

*3) Analysis of Parameter $\gamma$:* From Fig. 4(c), we can see that when the $\gamma$ value is greater than $10^0$, the TDAE algorithm achieves the optimal performance level. When it is less than $10^0$, the performance significantly tends to low level. The reason is that too small parameter values reduce the contribution of network's decoding, enlarging the difference between the output features of each layer and the input data, so that the output features are not conducive to classification calculations.

*4) Analysis of Parameter $\beta$:* It can be seen from Fig. 4(d) that the performance of the TDAE method is maximized when the value of parameter $\beta$ is $10^{-1}$. When the parameter value is less than $10^{-1}$, the performance of the algorithm decreases, as the usability of the auxiliary variable is weakened, leading to a larger deviation in the optimization output of the algorithm. When the parameter value is greater than $10^{-1}$, the performance of the algorithm also decreases, as the usability of the auxiliary variable becomes too strong, reducing the contribution of other components in the model.

### C. Algorithm Comparison

*1) Competing Methods:* We here compare the proposed TDAE method with eight state-of-the-art classification approaches including RAW-SVM [7], LRDLSR [17], DRDL [18], CNN-PPN [31], CDSAE [21], LeViT [36], HiT [37], and DRN [22]. The RAW-SVM uses the raw pixels of HSIs as discriminative features and the SVM classifier for label prediction. The LRDLSR and DRDL are, respectively, based on low-rank representation. The CDSAE and DRN are designed based on the architecture of deep autoencoder networks. The LeViT and HiT are transformer-based.

*2) Experimental Settings:* In the following experiments, we provide experimental settings related to algorithm comparison below: For the proposed TDAE method, the selection of parameter values can refer to the analysis results in Section III-B, that is, $\alpha = 1$, $\lambda = 0.1$, $\gamma = 1$, and $\beta = 0.1$. Regarding the eight competing methods, the corresponding parameter values can be optimally suggested by [7], [17], [18], [21], [22], [31], [36], and [37]. The comparison results and analysis details are presented as follows.

*3) Comparison Results:* As listed in Tables I–III, the quantitative metric results including OA, AA, and Kappa of RAW-SVM, LRDLSR, DRDL, CNN-PPN, CDSAE, LeViT, HiT, DRN, and our TDAE method on the Salinas, Trento,

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TABLE II
METRIC RESULTS OF DIFFERENT METHODS ON THE TRENTO DATASET

| Categories | Classification Methods | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | RAW-SVM | LRDLSR | DRDL | CNN-PPN | CDSAE | LeViT | HiT | DRN | TDAE |
| Apple trees | 55.56 | 73.36 | 97.53 | 95.54 | **99.72** | 89.46 | 91.50 | 99.66 | **99.72** |
| Buildings | 66.14 | 71.77 | 84.21 | 80.37 | 74.96 | 82.49 | 87.90 | 89.26 | **94.03** |
| Ground | 74.63 | 37.11 | 91.81 | 74.58 | 84.86 | 90.51 | 94.37 | 96.00 | **97.45** |
| Woods | 94.73 | 96.54 | 99.61 | 99.25 | **100** | 99.03 | **100** | 99.83 | 99.85 |
| Vineyard | 85.89 | 87.83 | 98.05 | 98.92 | 97.30 | 97.53 | 98.12 | **100** | 99.96 |
| Roads | 67.92 | 73.99 | 54.50 | 68.97 | 77.56 | 77.66 | 86.12 | 89.78 | **91.39** |
| OA(%) | 80.55 | 84.73 | 92.45 | 93.25 | 94.02 | 93.26 | 95.50 | 97.73 | **98.39** |
| AA(%) | 74.15 | 73.43 | 87.62 | 86.27 | 89.07 | 89.44 | 93.00 | 95.75 | **97.07** |
| Kappa(%) | 73.80 | 79.55 | 89.93 | 90.98 | 92.04 | 90.99 | 93.99 | 96.97 | **97.85** |

TABLE III
METRIC RESULTS OF DIFFERENT METHODS ON THE HOUSTON DATASET

| Categories | Classification Methods | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | RAW-SVM | LRDLSR | DRDL | CNN-PPN | CDSAE | LeViT | HiT | DRN | TDAE |
| Grass_healthy | 75.25 | 93.45 | 93.38 | 96.50 | 96.65 | 95.23 | 96.60 | 95.76 | **98.36** |
| Grass_stressed | 66.29 | 88.03 | 88.11 | 90.48 | 92.30 | 91.61 | 93.88 | 94.88 | **95.70** |
| Grass_synergie | 90.72 | 97.69 | 98.17 | 96.85 | **100** | 98.86 | 98.32 | 99.71 | **100** |
| Tree | 59.25 | 83.38 | 87.63 | 86.76 | 83.54 | 84.17 | 90.67 | 93.19 | **94.06** |
| Soil | 78.32 | 89.72 | 93.83 | 95.99 | 94.92 | 90.46 | 96.75 | 92.11 | **97.88** |
| Water | 78.23 | 95.57 | 90.32 | 96.59 | 92.90 | 96.24 | 97.37 | 96.55 | **98.52** |
| Residential | 61.27 | 84.18 | 86.81 | 88.66 | 86.31 | 83.60 | 88.70 | **92.43** | 91.36 |
| Commercial | 60.30 | 83.25 | 90.46 | 90.48 | 89.96 | 89.65 | 91.30 | 92.96 | **96.30** |
| Road | 64.20 | 83.91 | 87.24 | 87.50 | 87.90 | 86.18 | 90.27 | **91.92** | 90.18 |
| Highway | 72.87 | 95.43 | 94.38 | 98.24 | 94.94 | 95.27 | 97.03 | 97.22 | **97.72** |
| Railway | 81.39 | 96.81 | 93.15 | 96.81 | 97.06 | 95.70 | 97.02 | **97.18** | 96.33 |
| Parking lot 1 | 75.42 | 94.63 | 94.14 | 95.96 | 95.94 | 92.61 | 96.89 | 96.25 | **97.02** |
| Parking lot 2 | 78.02 | 93.93 | 94.24 | 94.21 | 95.89 | 90.78 | 96.13 | 94.46 | **98.02** |
| Tennis court | 84.63 | 92.51 | 96.81 | 98.12 | 99.22 | 97.52 | **100** | **100** | **100** |
| Running track | 78.84 | 95.82 | 95.66 | 96.62 | 96.74 | 98.40 | **100** | 99.42 | 98.43 |
| OA(%) | 71.88 | 90.43 | 91.73 | 93.40 | 92.97 | 91.58 | 94.72 | 95.11 | **96.06** |
| AA(%) | 73.67 | 91.22 | 92.29 | 93.98 | 93.62 | 92.42 | 95.40 | 95.60 | **96.66** |
| Kappa(%) | 69.59 | 89.66 | 91.06 | 92.87 | 92.40 | 90.90 | 94.29 | 94.71 | **95.74** |

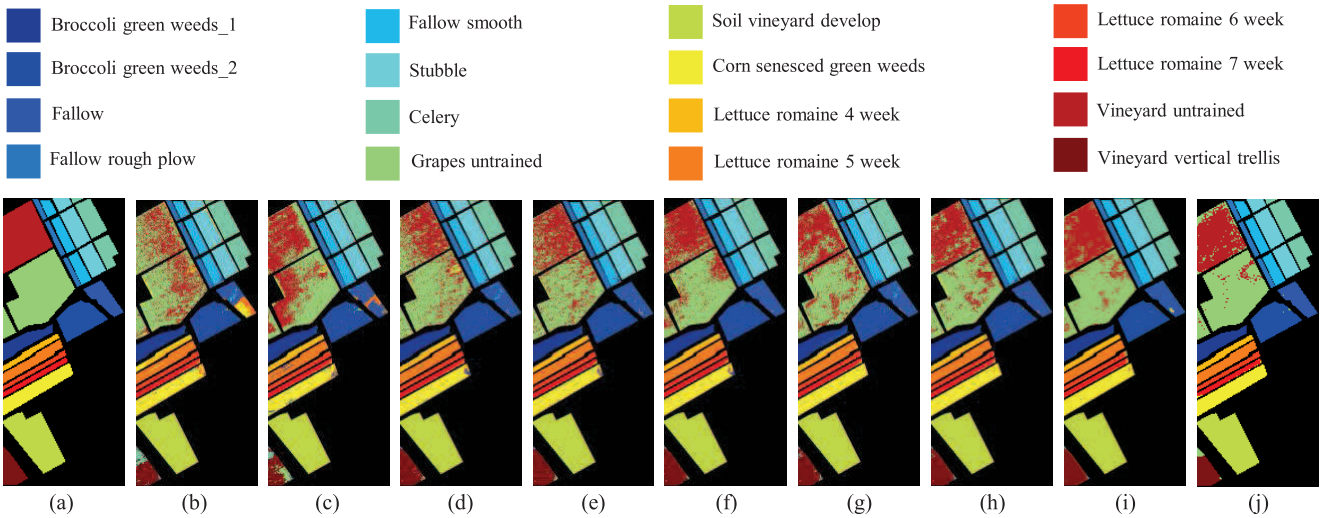



Fig. 5. Classification maps of different methods with 5% ratio of training samples on the Salinas data. (a) Ground truth. (b) RAW-SVM. (c) LRDLSR. (d) DRDL. (e) CNN-PPN. (f) CDSAE. (g) LeViT. (h) HiT. (i) DRN. (j) TDAE.

and Houston datasets are presented. Each listed value is the averaged result over five times and the best values have been marked in bold. From these three tables, it can be easily found that in most of the objective categories from the three

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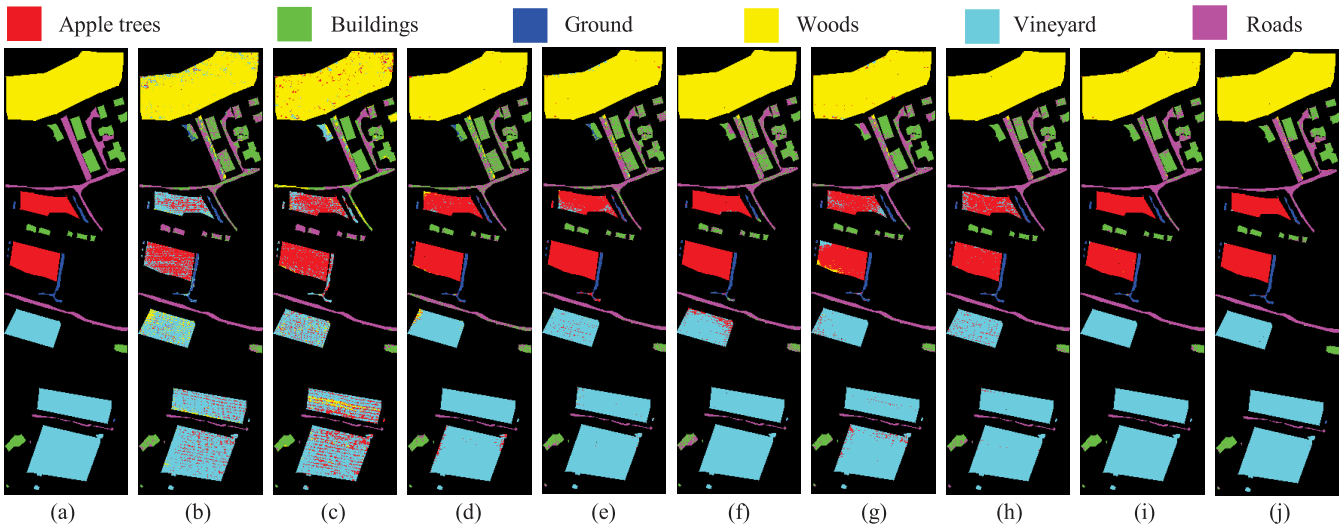



Fig. 6. Classification maps of different methods with 1% ratio of training samples on the Trento data. (a) Ground truth. (b) RAW-SVM. (c) LRDLSR. (d) DRDL. (e) CNN-PPN. (f) CDSAE. (g) LeViT. (h) HiT. (i) DRN. (j) TDAE.

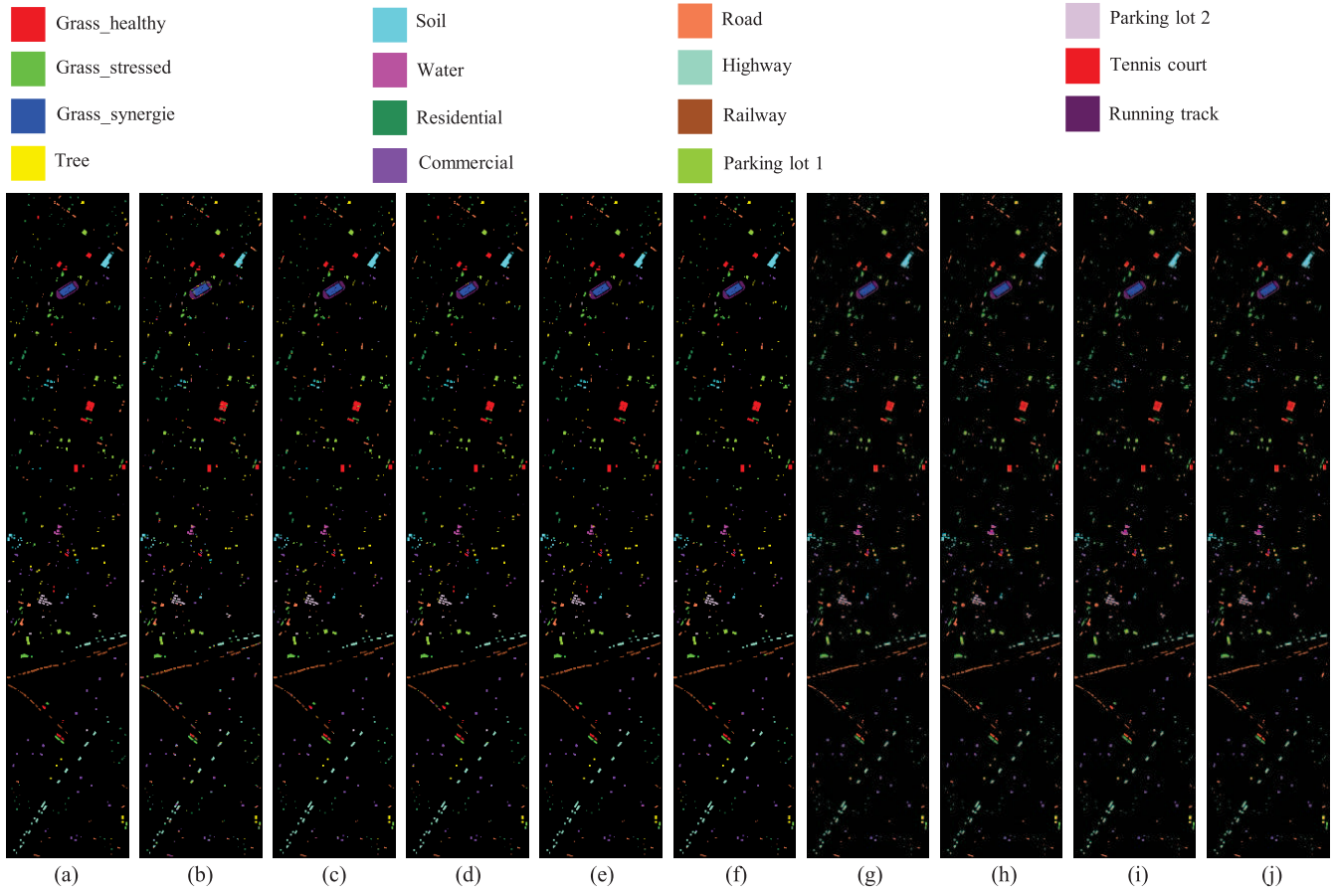



Fig. 7. Classification maps of different methods with 1% ratio of training samples on the Houston data. (a) Ground truth. (b) RAW-SVM. (c) LRDLSR. (d) DRDL. (e) CNN-PPN. (f) CDSAE. (g) LeViT. (h) HiT. (i) DRN. (j) TDAE.

datasets, the classification accuracy of our proposed TDAE method is higher than that of the eight competing approaches. In addition, the proposed TDAE method achieves the best results in OA, AA, and Kappa on the Salinas, Trento, and Houston datasets.

Furthermore, we also give the classification maps of RAW-SVM, LRDLSR, DRDL, CNN-PPN, CDSAE, LeViT, HiT, DRN, and our proposed TDAE method on the Salinas, Trento, and Houston datasets in Figs. 5–7. From these figures, it can be observed that the TDAE method obtains the classification

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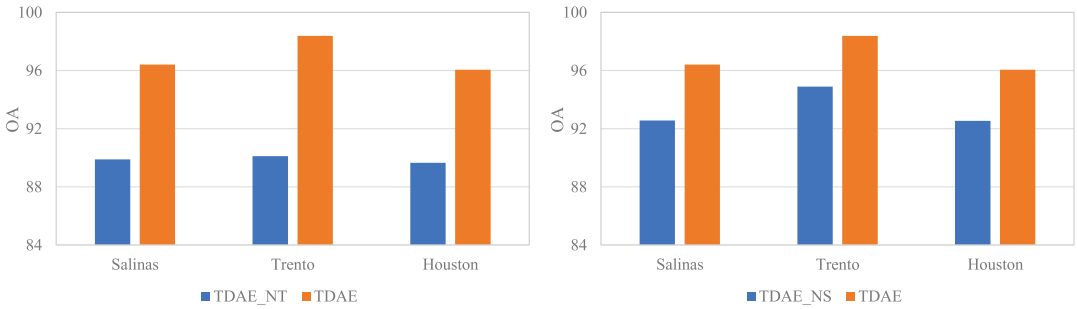



Fig. 8. TDAE is compared with the form without the tensor decomposition item (denoted by TDAE_NT) and sparse regularization item (denoted by TDAE_NS). OA is used to measure the results.

maps that are closer to the ground-truth maps compared with competing approaches on the datasets. For example, in Fig. 5, the "Grapes unstrained" category yielded by the eight competing approaches in Fig. 5(b)–(i) contains lots of classification errors. However, such errors are significantly mitigated in the output map [see Fig. 5(j)] exported by our TDAE method. In Fig. 6, we easily find the clear misclassification results on the "Apple trees" category yielded via the eight competing approaches in Fig. 6(b)–(i), but which are greatly improved by our TDAE method in Fig. 6(j). In Fig. 7, it can be observed that the classification maps for the competing approaches in Fig. 7(b)–(i) have more difference with the ground truth in Fig. 7(a) than the TDAE method in Fig. 7(j).

According to the above qualitative and quantitative evaluation, it is concluded that the proposed TDAE method outperforms the RAW-SVM, LRDLSR, DRDL, CNN-PPN, CDSAE, LeViT, HiT, and DRN. The main reason is that unlike these competing approaches, our TDAE method is constructed on the deep neural network with hierarchical representation, which can concurrently use both deep and shallow features for HSI classification by performing network tensorization.

### D. Ablation Study

For sake of comprehensively exploring the performance of the proposed TDAE method, we conduct the ablation study to investigate the effectiveness of the tensor decomposition item and sparse regularization item included in model (2). We here use the OA index to measure the results, which are visually presented in Fig. 8. The details of the ablation study are shown as follows.

*1) Effectiveness of Tensor Decomposition Item:* The tensor decomposition item in (2) is introduced to achieve the capacity of hierarchical representation for the proposed TDAE method, which assists to concurrently use both deep and shallow features so as to improve the classification performance of HSIs. To validate effectiveness of the tensor decomposition item, the TDAE method is here compared with the form without this item (denoted by TDAE_NT) on the Salinas, Trento, and Houston datasets, with the results shown in Fig. 8(a). As can be clearly seen from this figure, the TDAE generates the larger OA values than TDAE_NT. In other words, when the tensor decomposition item is removed from the TDAE, the capacity of hierarchical representation will disappear and the classification performance will seriously become vestigial. Therefore, we conclude great effectiveness of the tensor decomposition item in (2).

*2) Effectiveness of Sparse Regularization Item:* The sparse regularization item is used to constrain the sparsity of all the layers' outputs of our TDAE network form, which guarantees the extracted feature to be well discriminative. For sake of exploring the effectiveness of the sparse regularization item, the TDAE is compared with the form without the item (denoted by TDAE_NS) on the Salinas, Trento, and Houston datasets, which is experimentally presented in Fig. 8(b). As can be directly found, the OA values of TDAE are bigger than those of TDAE_NS. That is, once the sparse regularization item is removed from the TDAE, the classification accuracy will decrease, which demonstrates that this item plays an important role in the proposed TDAE method.

## IV. Conclusion

In this study, we propose a novel TDAEs' network for the task of HSI classification. More specifically, a deep autoencoders' network with a fully connection layer is first introduced as the basic framework. Second, we build the tensor decomposition constraint item, which is further introduced into the basic framework. This not only achieves the integration of multilayer output features but also captures the structure information among outputs. With such manner, the network's ability for hierarchical representation is significantly enhanced. Third, we theoretically explore the relationship between the tensor decomposition and matrix form, and based on this, we further design an alternating update optimization scheme to obtain the solutions of the constructed model. Finally, when the optimal solutions are obtained, the desired feature forms will be extracted, which are input into the fully connection layer for label generation of the given HSI. We evaluate the performance of the proposed TDAE method on the Salinas, Trento, and Houston datasets, with the experimental results validating its effectiveness and superiority in HSI classification.

It should be noted that in the proposed TDAE method for HSI classification, the fully connected layer based on soft-max is directly used to classify the features, which does not consider the correlation between the feature extraction and the soft-max classifier. In other words, the design of the

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TDAE method is independent of the importance of the softmax classifier, which may limit the generalization. In the future work, we will develop the feature extraction strategy with the importance of various classifiers, for the sake of improving the generalization.

## REFERENCES


[1] S. Li, R. Dian, L. Fang, and J. M. Bioucas-Dias, "Fusing hyperspectral and multispectral images via coupled sparse tensor factorization," *IEEE Trans. Image Process.*, vol. 27, no. 8, pp. 4118–4130, Aug. 2018.

[2] R. Dian, S. Li, L. Fang, T. Lu, and J. M. Bioucas-Dias, "Nonlocal sparse tensor factorization for semiblind hyperspectral and multispectral image fusion," *IEEE Trans. Cybern.*, vol. 50, no. 10, pp. 4469–4480, Oct. 2020.

[3] R. A. Borsoi, T. Imbiriba, J. C. M. Bermudez, and C. Richard, "A blind multiscale spatial regularization framework for kernel-based spectral unmixing," *IEEE Trans. Image Process.*, vol. 29, pp. 4965–4979, 2020.

[4] J. Han, D. Zhang, G. Cheng, L. Guo, and J. Ren, "Object detection in optical remote sensing images based on weakly supervised learning and high-level feature learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 6, pp. 3325–3337, Jun. 2015.

[5] L. Fang, S. Li, W. Duan, J. Ren, and J. A. Benediktsson, "Classification of hyperspectral images by exploiting spectral–spatial information of superpixel via multiple kernels," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 12, pp. 6663–6674, Dec. 2015.

[6] J. Jiang, J. Ma, Z. Wang, C. Chen, and X. Liu, "Hyperspectral image classification in the presence of noisy labels," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 851–865, Feb. 2019.

[7] L. He, J. Li, C. Liu, and S. Li, "Recent advances on spectral–spatial hyperspectral image classification: An overview and new guidelines," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 3, pp. 1579–1597, Mar. 2018.

[8] B. Roh, W. Shin, I. Kim, and S. Kim, "Spatially consistent representation learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 1144–1153.

[9] Y. Bengio, A. Courville, and P. Vincent, "Representation learning: A review and new perspectives," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 35, no. 8, pp. 1798–1828, Aug. 2013.

[10] B. Schölkopf et al., "Toward causal representation learning," *Proc. IEEE*, vol. 109, no. 5, pp. 612–634, May 2021.

[11] C. Xing, M. Wang, C. Dong, C. Duan, and Z. Wang, "Using Taylor expansion and convolutional sparse representation for image fusion," *Neurocomputing*, vol. 402, pp. 437–455, Aug. 2020.

[12] C. Xing, M. Wang, C. Dong, C. Duan, and Z. Wang, "Joint sparse-collaborative representation to fuse hyperspectral and multispectral images," *Signal Process.*, vol. 173, Aug. 2020, Art. no. 107585.

[13] Y. Chen, N. M. Nasrabadi, and T. D. Tran, "Hyperspectral image classification using dictionary-based sparse representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 10, pp. 3973–3985, Oct. 2011.

[14] X. Zhang, P. Weng, J. Feng, E. Zhang, and B. Hou, "Spatial–spectral classification based on group sparse coding for hyperspectral image," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jan. 2013, pp. 1745–1748.

[15] H. Zhang, J. Li, Y. Huang, and L. Zhang, "A nonlocal weighted joint sparse representation classification method for hyperspectral imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2056–2065, Jun. 2017.

[16] Z. He, L. Liu, R. Deng, and Y. Shen, "Low-rank group inspired dictionary learning for hyperspectral image classification," *Signal Process.*, vol. 120, pp. 209–221, Mar. 2016.

[17] Z. Chen, X.-J. Wu, and J. Kittler, "Low-rank discriminative least squares regression for image classification," *Signal Process.*, vol. 173, Aug. 2020, Art. no. 107485.

[18] C. Xing, M. Wang, Z. Wang, C. Duan, and Y. Liu, "Diagonalized low-rank learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5507812.

[19] Y. Chen, Z. Lin, X. Zhao, G. Wang, and Y. Gu, "Deep learning-based classification of hyperspectral data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2094–2107, Jun. 2014.

[20] Y. Chen, X. Zhao, and X. Jia, "Spectral–spatial classification of hyperspectral data based on deep belief network," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 6, pp. 2381–2392, Jun. 2015.

[21] P. Zhou, J. Han, G. Cheng, and B. Zhang, "Learning compact and discriminative stacked autoencoder for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 7, pp. 4823–4833, Jul. 2019.

[22] C. Xing, J. Zhao, Z. Wang, and M. Wang, "Deep ring-block-wise network for hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 10, pp. 14125–14137, Oct. 2024.

[23] Y. Xu, B. Du, F. Zhang, and L. Zhang, "Hyperspectral image classification via a random patches network," *ISPRS J. Photogramm. Remote Sens.*, vol. 142, pp. 344–357, Aug. 2018.

[24] M. Zhang, W. Li, and Q. Du, "Diverse region-based CNN for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 27, no. 6, pp. 2623–2634, Jun. 2018.

[25] J. He, L. Zhao, H. Yang, M. Zhang, and W. Li, "HSI-BERT: Hyperspectral image classification using the bidirectional encoder representation from transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 165–178, Sep. 2019.

[26] O. Okwuashi and C. E. Ndehedehe, "Deep support vector machine for hyperspectral image classification," *Pattern Recognit.*, vol. 103, Jul. 2020, Art. no. 107298.

[27] B. Xi, J. Li, Y. Li, R. Song, and Q. Du, "Deep prototypical networks with hybrid residual attention for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 3683–3700, Jun. 2020.

[28] A. Sellami, A. Ben Abbes, V. Barra, and I. R. Farah, "Fused 3-D spectral–spatial deep neural networks and spectral clustering for hyperspectral image classification," *Pattern Recognit. Lett.*, vol. 138, pp. 594–600, Oct. 2020.

[29] C. Zheng and N. Wang, "Collaborative representation with k-nearest classes for classification," *Pattern Recognit. Lett.*, vol. 117, pp. 30–36, Jan. 2019.

[30] X. Cao, J. Yao, Z. Xu, and D. Meng, "Hyperspectral image classification with convolutional neural network and active learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, pp. 4604–4616, Jul. 2020.

[31] W. Li, G. Wu, F. Zhang, and Q. Du, "Hyperspectral image classification using deep pixel-pair features," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 2, pp. 844–853, Feb. 2017.

[32] L. Fang, G. Liu, S. Li, P. Ghamisi, and J. A. Benediktsson, "Hyperspectral image classification with squeeze multibias network," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 3, pp. 1291–1301, Mar. 2019.

[33] Y. Shen et al., "Efficient deep learning of nonlocal features for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 6029–6043, Jul. 2021.

[34] Y. Xu, B. Du, and L. Zhang, "Robust self-ensembling network for hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 3, pp. 3780–3793, Mar. 2024.

[35] M. Jaderberg, K. Simonyan, A. Zisserman, and K. Kavukcuoglu, "Spatial transformer networks," in *Proc. Adv. Neural Inf. Process. Syst. (NIPS)*, vol. 28, Dec. 2015, pp. 2017–2025.

[36] B. Graham et al., "LeViT: A vision transformer in ConvNet's clothing for faster inference," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 12239–12249.

[37] X. Yang, W. Cao, Y. Lu, and Y. Zhou, "Hyperspectral image transformer classification networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5528715.

[38] L. Scheibenreif, M. Mommert, and D. Borth, "Masked vision transformers for hyperspectral image classification," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2023, pp. 2166–2176.

[39] C. Xing, Y. Cong, C. Duan, Z. Wang, and M. Wang, "Deep network with irregular convolutional kernels and self-expressive property for classification of hyperspectral images," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 12, pp. 10747–10761, Mar. 2023.

[40] B. Yang, J. Chen, and M. Ye, "Shallow-deep collaborative learning for unsupervised visible-infrared person re-identification," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 16870–16879.

[41] K. Li, C. Zou, S. Bu, Y. Liang, J. Zhang, and M. Gong, "Multi-modal feature fusion for geographic image annotation," *Pattern Recognit.*, vol. 73, pp. 1–14, Jan. 2018.

[42] G. H. Golub and C. F. Van Loan, *Matrix Computations*. Baltimore, MD, USA: The Johns Hopkins Univ. Press, 2013.

[43] O. Axelsson, *Iterative Solution Methods*. Cambridge, U.K.: Cambridge Univ. Press, 1996.


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